
Structure, Not Confidence: A Four-Way Benchmark for Deciding When to Act, Ask, Refuse, or Defer

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Abstract

An assistant that cannot tell an under-specified request from a well-specified one will answer both, and one of those answers is a guess dressed as an answer. We ask whether a model can recognise that it lacks the information to act safely, and choose among *acting*, *asking*, *refusing*, and *deferring* according to the ambiguity and the stakes. We build CARB, an 840-item four-way benchmark whose gold labels are inherited from the source annotations of CoCoNOT, CLAMBER, and IN3 rather than invented, and run five elicitation regimes over identical items on five models. Three results follow. First, *structure beats confidence*: surfacing a typed action ontology reaches 0.60–0.72 four-way accuracy against 0.23 uniform-random, and beats an optimally-routed verbalised confidence by 0.16–0.28 on every model (all $p_{\text{Holm}} < 10^{-8}$); the scalar separates act from not-act (AUROC 0.79–0.87) but is at chance at separating *why* not (DEFER vs. REFUSE AUROC 0.44–0.56). Second, the failure is specific: models judge “is this safe?” at 0.89 and “can I do this?” at 0.94, but “was I told enough?” at 0.57. Third, announcing high stakes moves the decision criterion ($\Delta c = -0.44$, $p < 0.001$) without improving discrimination ($\Delta d' = -0.11$, $p = 0.53$), and degrades it outright under a plain ask-affordance. LoRA fine-tuning an open 4B model reaches 0.794 in-distribution but 0.470 out-of-source, where a frozen probe on the untrained model already reaches 0.779. Once errors are priced, every prompted frontier policy is worth less than always asking once a wrong action costs 5–10 $\times$ a needless refusal.

1 Introduction

Ask a deployed assistant to “book me a flight to Springfield” and it will book one. There are more than thirty Springfields in the United States. The assistant did not lack the capability to ask which one; it lacked the judgment that asking was the right move. This is the failure mode we study: not that models cannot act, but that they act when they should have said something else.

The right response to a request is not always an answer, and not always a refusal either. A request can be under-specified in a way the user can fix in one turn — then the model should **ask**. It can be something that should not be done at all, or that has no determinate answer — then the model should **refuse**. It can be blocked by the model’s own missing capability, modality, or access — then the model should **defer** and hand off. Only in the fourth case should it **act**. These four are genuinely different: a refusal and a clarifying question have opposite effects on a conversation, and a binary benchmark that scores both as “did not answer” cannot tell them apart.

Why this matters now. Assistants are being wired to tools that write to databases, send messages, and execute code, where acting on a misread intent is not a wrong answer but a wrong action. The stated hypothesis in this line of work — that a model can be trained or prompted to recognise when it lacks sufficient information, *by estimating its uncertainty*, and choose among the four actions

depending on the stakes — is the design assumption behind a large amount of deployed scaffolding. It has never been tested as a whole.

The gap. Three literatures each test one boundary at a time: clarification research asks *ask vs. answer* [Zhang and Choi, 2023, Zhang et al., 2024], abstention research asks *answer vs. decline* [Wen et al., 2024], and agentic help-seeking asks *escalate vs. continue* [Trinh et al., 2026]. The few papers that propose the full action space are small or leave the hard part open: Unlu [2026] runs $n=32$ items on a single model inside one context window and concedes its numbers are upper bounds; Baan et al. [2026] reports that separating clarify from abstain “remains challenging”; Su and Cardie [2026] shows models recognise ambiguity and answer anyway but does not decompose *which* recognition fails. Nobody has run the head-to-head between typed structure and scalar uncertainty at scale, and no paper in this corpus reports a clean stakes $\times$ ambiguity factorial.

Our approach. We build CARB (Clarification–Action Routing Benchmark), 840 items whose gold four-way labels are inherited from the source datasets’ own annotations through a fixed, published mapping — no labels are invented. We then hold the items fixed and vary only *how the decision is elicited*, across five regimes: the bare request, the request plus a bare ask-affordance, a typed action ontology with an ordered decision procedure, a verbalised confidence that we route into actions with the best cross-validated router that can be fitted to it, and four isolated property judgments that never ask the model what to do (figure 1). Every model sees every item under every regime, so every comparison is within-item. We run four frontier models and one open-weight model, and on the open model we add LoRA fine-tuning and a frozen linear probe.

What we find. Typed prompting reaches 0.60–0.72 four-way accuracy against 0.231 uniform-random and 0.285 best-single-action, and cuts over-commitment (acting on items that should have been withheld) from 45–58% to 9–25%. It beats optimally-routed scalar confidence by 0.158–0.275 accuracy on every model (Cohen’s h 0.32–0.57, all $p_{\text{Holm}} < 10^{-8}$). The scalar is not merely mis-routed: it is at chance at separating DEFER from REFUSE (AUROC 0.436–0.563), and adding it to a typed policy never helps. Recognition beats behaviour, but the property that fails is exactly the one the hypothesis names — “is the information sufficient?” is 0.565 on items that needed a question, against 0.893 for safety and 0.943 for capability. Announcing high stakes shifts the criterion by -0.44 ($p < 0.001$) with no gain in d' , and the ambiguity $\times$ stakes interaction is $\beta = +0.010$ ($p = 0.97$). LoRA SFT on an open 4B model lifts typed routing from 0.577 to 0.794 in-distribution but scores 0.470 out-of-source, below its own untrained prompted baseline, while a frozen probe on the base model already reaches 0.779. Priced under an error-cost model, every frontier model’s best prompted policy falls below the trivial always-ask policy once a wrong action costs 5–10 $\times$ a needless refusal.

- We release CARB, an 840-item four-way action-routing benchmark with source-inherited labels, a contrast set for over-refusal, an out-of-source transfer split, and a published label-mapping table with sensitivity analyses over three alternative mappings.
- We run the first at-scale head-to-head between typed decision structure and scalar uncertainty on identical items across five models, and show structure wins everywhere by 0.13–0.28 accuracy while the scalar is provably at chance on the distinction that matters.
- We decompose the recognition–behaviour gap into four property judgments and localise the failure to *information sufficiency*, and we show the same two patterns reproduce on an open-weight model, so they are not an artefact of one lab’s post-training.
- We run a stakes $\times$ ambiguity factorial with a paired bootstrap of the change in signal-detection parameters, and show announced stakes buy indiscriminate caution, not calibrated caution.
- We price every policy under an error-cost model and show that all prompted frontier policies lose to always-asking at moderate error costs — the deployment-relevant bar that accuracy hides.

Section 2 positions the work, section 3 describes CARB and the regimes, section 4 reports the four experiments, and sections 5 and 6 discuss limits and implications.

2 Related Work

Clarification: when to ask. Zhang and Choi [2023] give the canonical decomposition — when to clarify, what to ask, and how to use the answer — and propose INTENT-SIM, which scores ambiguity

by the entropy of simulated user intents. Their evaluation protocol, AUROC plus performance under a fixed interaction budget, is the one we adopt in section 4 for the scalar regime. Their absolute numbers are also a caution: AUROCs cluster at 0.50–0.63, and SELF-ASK scores 0.371 on machine translation — below chance. Zhang et al. [2024] supply a balanced ask/don’t-ask benchmark with a taxonomy, which we use as our out-of-source transfer split precisely because it was never used for tuning. Qian et al. [2024] annotate the *importance* of each missing detail, which gives us item-level stakes that nobody announces to the model. Unlike this line of work, we do not ask whether to clarify; we ask which of four actions is right, because *don’t clarify* is three different things.

Abstention: when to decline. Wen et al. [2024] survey abstention along query, model, and human-value axes and argue it should be treated as a meta-capability rather than a per-task trick. ABSTENTIONBENCH [AbstentionBench Authors, 2025] reports that reasoning fine-tuning *hurts* abstention by roughly 24%, and ships a human-validated binary abstention judge whose prompt we extend into a four-way judge (section 3). The limitation we inherit from this literature is exactly the one it is built on: abstention is binary, so refusing and deferring score identically.

Agentic help-seeking. HiL-BENCH [Trinh et al., 2026] places human-validated blockers in SWE-BENCH PRO and BIRD tasks that surface only through execution, and finds frontier models score 75–89% pass@3 with full information but 4–24% when they must judge whether to ask — despite having an `ask_human()` tool. Their ASK-F1 metric, the harmonic mean of question precision and blocker recall, is spam-resistant by construction and we report it throughout. They also show RLVR on shaped ASK-F1 transfers across domains, which is the strongest existing evidence for the trainability clause; our section 4 training arm supervises the typed label instead, and finds transfer does *not* follow for free. Wang et al. [2024] independently find that agents fabricate missing tool arguments rather than asking, which is the mechanism behind the over-commitment rates we measure.

The full action space. Two recent papers propose ontologies close to ours. Unlu [2026] define four support states by minimal repair distance — complete, clarifiable, support-blocked, unsupported-now — and report that scalar confidence prompting avoids over-commitment but collapses the three-way deferral space to 58.3% typed accuracy, while surfacing the categorical ontology reaches 91.7%. The result is the direct ancestor of our Experiment 2, but it rests on 32 items, one model, and a dual-persona audit run inside a single context window, which the author concedes yields upper-bound estimates. We replicate the *direction* at $15\times$ the sample size, across five models, with contexts isolated — and find the 91.7% figure was indeed an upper bound (we see 0.650–0.764 typed-deferral accuracy). Baan et al. [2026] let a model reason over its own sampled belief state to pick answer, clarify, or abstain, and name clarify-versus-abstain separation as the open problem; our confusion matrices quantify it (26% of ASK items routed to REFUSE under typed prompting). SAFETY SENTRY [Safety Sentry Authors, 2026] routes over execute/ask/refuse with a single decoding-time threshold repositioned per deployment risk — the cleanest existing mechanism for the stakes clause, and the motivation for measuring criterion shift separately from discrimination.

Recognition vs. behaviour. Su and Cardie [2026] show models identify ambiguity when explicitly asked to judge it, yet default to direct answers in ordinary QA, and that retrieved context widens the gap. This is the framing of our Experiment 1. Building on it, we do not stop at showing the gap exists: we decompose recognition into four separately-scored properties and show that safety and capability recognition are near-ceiling while information-sufficiency recognition is near chance, which changes what an intervention should target.

Uncertainty estimation. Semantic entropy over NLI-clustered samples [Kuhn et al., 2023] is the standard strong baseline, and conformal methods [Yadkori et al., 2024] give distribution-free risk control at a chosen level. Selective Prediction Authors [2026] report a negative result for entropy-based selective prediction. Zong et al. [2026] announce answer/abstain payoff matrices in the prompt, which we adapt directly for our announced-stakes manipulation. Our contribution here is negative and geometric: we show that the problem with a scalar is not its sharpness but its dimensionality — three deferral types cannot be recovered from an ordering of one number, and empirically the models do not even order them consistently.

Evaluation beyond a single turn. RegretBench Authors [2026] score clarification as a *policy* under hidden intent, decomposing regret against a reference policy into intent resolution, interaction cost, and ineffective clarification, and Wu and Fard [2024] measure the cost of a wrong assumption objectively by running the resulting code. Both are the natural next step past what we measure: CARB

Action	Who is blocked	Meaning	Source annotation
ACT	nobody	well specified, safe, within capability	CoCoNOT <i>contrast</i> set; CLAMBER <i>require_clarification</i> =0; IN3 <i>vague</i> =false
ASK	the user	missing or ambiguous, and the user can fix it in one turn	CoCoNOT <i>incomplete requests</i> ; CLAMBER =1; IN3 <i>vague</i> =true
REFUSE	nobody can	should not be done at all — harmful, or no determinate answer exists	CoCoNOT <i>safety concerns, humanizing, indeterminate</i>
DEFER	the model	the model lacks the capability, modality, or access; hand off	CoCoNOT <i>unsupported requests</i>

Table 1: The CARB action space and its source-inherited labels. No gold label is authored by us: each item carries its source dataset’s own annotation through a fixed mapping table published with the benchmark. The middle column is the operative distinction — ASK, REFUSE, and DEFER differ in *who* is blocked, which is exactly what a binary answer/abstain benchmark cannot express.

scores a single routing decision per request, so it can say whether asking was right but not whether the conversation that followed was efficient.

3 CARB: benchmark, regimes, and protocol

3.1 Problem formulation

A request x is routed to one of four actions $a \in \{\text{ACT}, \text{ASK}, \text{REFUSE}, \text{DEFER}\}$. The gold action is determined by three properties of the request, in a fixed order:

- REFUSE if the request is unsafe or has no determinate answer — it should not be done at all;
- DEFER if it is blocked by the *model*’s missing capability, modality, or access — hand off;
- ASK if information is missing or ambiguous and the *user* can supply it in one turn;
- ACT otherwise: well specified, safe, and within capability.

The ordering matters. A request that is both unsafe and under-specified is a REFUSE, not an ASK, and asking the user to specify it would be the wrong move. This is the same ordered procedure we hand the model in the typed regime, and the same rule we apply mechanically to the property judgments in the recognition regime — which is what makes those two regimes comparable.

3.2 Benchmark construction

Existing benchmarks are binary. CARB assembles a four-way action space from sources that already carry the distinction [Brahman et al., 2024, Zhang et al., 2024, Qian et al., 2024], and **never invents a gold label**: every item inherits its label from its source dataset’s own annotation through a fixed table published with the benchmark (table 1).

CARB contains **840 items**: a dev split of 160 (threshold tuning only), a **test split of 480** on which every headline number is computed, and a transfer split of 200 drawn from CLAMBER, a source never used for tuning or training. Sampling is balanced by gold action over the 640-item core pool (280 ASK/ 280 ACT/ 140 REFUSE/ 140 DEFER); the dev/test cut is random, so the test split is 137 ASK/ 129 ACT/ 112 REFUSE/ 102 DEFER. Two mapping cells are flagged *contested* and every headline number is recomputed without them, under an alternative mapping that sends false-presupposition items to REFUSE, and under a capability re-audit of the IN3 items (section 4.8). Training data for the fine-tuning arm (2,800 items) comes from CoCoNOT’s *train* split and AMBIGQA [Min et al., 2020]; prompt-level overlap with test is 0/479 and with transfer 3/200.

3.3 Elicitation regimes

The independent variable is how the decision is elicited, not what is being decided (figure 1). Every model sees every item under all five regimes, so all comparisons are within-item and paired.

Two design choices deserve emphasis. First, **the scalar regime is given every advantage**: we do not pick a threshold by hand but search all 3-cut $\times$ 24-permutation routers — every way of slicing the confidence axis into four ordered bins and assigning the four actions to them — and select the best one under 5-fold cross-validation, so no item is ever predicted by a router that saw it. If a scalar confidence can support four-way routing at all, this procedure finds

Regime	What the model is shown	What it returns	What it measures
R0-DIRECT	the bare request	free text	default behaviour
R1-AFFORD	request + "you may answer, ask, decline, or explain a limitation"	free text	behaviour with an ask-affordance
R2-TYPED	request + four-action ontology + ordered decision procedure	JSON action + reason	typed categorical routing
R3-SCALAR	request + "how confident are you that you can safely do this now, 0-100?"	a number	uncertainty estimation
R4-RECOG	request + four yes/no property questions; <i>never</i> what to do	four booleans	recognition, isolated from behaviour

Figure 1: The five elicitation regimes. The items are identical across regimes and every model sees every item under all five, so all comparisons are within-item. R3-SCALAR’s number is turned into an action by the best $3\text{-cut} \times 24\text{-permutation}$ router fittable to it under 5-fold cross-validation; R4-RECOG’s booleans are turned into an action by a fixed rule structurally identical to the procedure R2-TYPED hands the model. R2-TYPED vs. R4-RECOG therefore isolates *who applies the rule*, and R2-TYPED vs. R3-SCALAR isolates *structure vs. a number*.

it. Second, **the recognition regime never asks the model what to do**: it asks four yes/no property questions (`information_sufficient`, `user_can_resolve`, `safe_and_determinate`, `within_capability`), and a fixed rule converts the booleans into an action. The typed and recognition regimes therefore differ only in *who applies the decision rule*, the model or us.

3.4 Models, judge, and statistics

Models. GPT-5, CLAUDE-SONNET-4.5, GEMINI-2.5-FLASH, and LLAMA-3.3-70B through OpenRouter at temperature 0, and QWEN3-4B locally (bf16, greedy decoding, thinking disabled) for the training and probing arm. The main run is 7,668 API calls at \$50.79; local compute is $4 \times$ NVIDIA RTX A6000 (48 GB).

Judge. Free-text responses (R0-DIRECT, R1-AFFORD) must be classified into the four actions. We extend ABSTENTIONBENCH’s human-validated binary abstention judge prompt to four ways and run it locally on QWEN3-14B by *constrained choice*: the judge prompt is completed with `{"behaviour": " "}` and the four action words are compared at that slot in a single forward pass. This makes format errors impossible and the decision deterministic. We check reliability two ways: against an independent second judge (QWEN3-4B) on 3,768 items, raw agreement 0.863 and Cohen’s $\kappa = 0.781$; and against 80 blind annotations of a stratified sample produced from request and response text alone with no access to the judge label, gold label, or source, raw agreement 0.812 and $\kappa = 0.750$. Both fall in the conventional substantial-agreement band. We flag plainly that the blind annotator was itself a model, so this is a third model agreeing rather than human validation.

Metrics. We report four-way accuracy; macro-F1; **ASK-F1** [Trinh et al., 2026], the harmonic mean of ask precision and ask recall, which no ask-everything policy can game; **over-commitment**, the fraction of non-ACT items acted on anyway [Unlu, 2026]; **contrast compliance**, accuracy on the contrast set of superficially similar but genuinely answerable requests, which no refuse-everything policy can game; and **typed-deferral accuracy**, the fraction of correctly-withheld items given the right *kind* of withholding. ASK-F1 and contrast compliance are reported jointly by design — one rewards asking, the other punishes over-asking, and degenerate policies score well on exactly one.

Baselines. Always-ACT, always-ASK (the majority class), always-REFUSE, uniform random, and a TF-IDF plus logistic-regression classifier trained on a disjoint COCONOT pool as a lexical-shortcut probe.

Statistics. Within-item comparisons use **McNemar’s exact test**, Holm-corrected across a pre-registered family of 24 comparisons. Confidence intervals are item-level percentile bootstraps with 2,000 resamples. Effect sizes are Cohen’s h . For the stakes experiment we bootstrap the *change* in signal-detection parameters between frames, resampling items and carrying each resampled item’s responses under both frames, because comparing two separately bootstrapped intervals is not a test. Seed 42 throughout.

3.5 Training and probing

On QWEN3-4B we add two supervised conditions. **LoRA SFT** [Hu et al., 2022]: 33M trainable parameters, 2 epochs on 2,576 prompt→action pairs disjoint from the test set, under the same typed prompt used for prompting, so the comparison isolates training from ontology. **Frozen linear probe**: a logistic regression on the last hidden state at the final prompt token of the *base* model — no fine-

Policy	Split	Acc	Macro-F1	ASK-F1	Over-commit ↓	Contrast compl.
always ACT	test	0.269	0.106	0.000	1.000	1.000
always ASK (majority)	test	0.285	0.111	0.444	0.000	0.000
always REFUSE	test	0.233	0.095	0.000	0.000	0.000
uniform random	test	0.231	0.230	0.243	0.254	0.240
TF-IDF + logistic regression	test	0.702	0.701	0.613	0.077	0.612
TF-IDF + logistic regression	transfer	0.355	0.196	0.574	0.110	0.130

Table 2: Reference policies on CARB. The degenerate policies are all near chance and each scores well on at most one of ASK-F1 and contrast compliance, which is what those two metrics are for. The TF-IDF probe is the important row: a bag-of-words classifier reaches **0.702** in-source, higher than every frontier model’s typed routing except CLAUDE-SONNET-4.5’s 0.721, and collapses to 0.355 out-of-source. Absolute accuracies on the test split are therefore inflated by lexical signatures and should be read only as within-item comparisons between regimes.

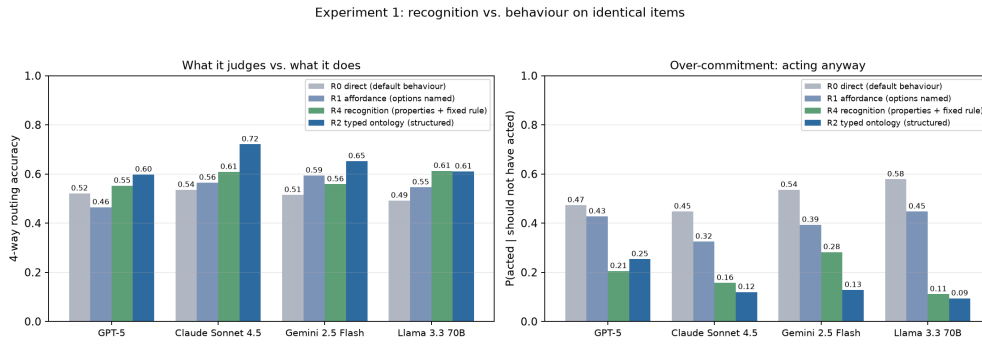


Figure 2: The recognition–behaviour gap. For every model, the action implied by the model’s own property judgments (R4-RECOG), applied through a fixed rule, is more accurate than what the model actually does under a plain prompt (R0-DIRECT). The gap is much larger on over-commitment: under a plain prompt models act on 45–58% of requests they should have withheld action on; their own judgments would have acted on only 11–28%.

tuning, no generation, just the model’s representation of a request it is about to answer. The probe answers a question prompting cannot: is the routing signal absent from the model, or present but unread?

4 Results

4.1 The benchmark is neither trivial nor free of shortcuts

Table 2 sets the reference points. Uniform random scores 0.231 and the best single-action policy 0.285, so four-way accuracy is a meaningful scale. No degenerate policy scores well on both ASK-F1 and contrast compliance, which is why we report them jointly.

We also report the shortcut we found rather than burying it. A TF-IDF plus logistic-regression classifier trained on a disjoint CoCoNOT pool reaches **0.702** on the test split — higher than every frontier model’s typed routing except CLAUDE-SONNET-4.5’s 0.721. CoCoNOT’s categories carry strong lexical signatures (“every”, “all ... that exists”, “current”, “I cannot”), and the same classifier collapses to 0.355 on the out-of-source transfer split. **Absolute accuracies on the test split are therefore inflated and should be read as within-benchmark comparisons between regimes on identical items, never as capability estimates.** Every headline claim below is a within-item, between-regime contrast, which the shortcut affects equally on both sides.

Model	Regime	Acc [95% CI]	Macro-F1	ASK-F1	Over-commit ↓	Contrast compl.	Typed-deferral
CLAUDE-SONNET-4.5	R0-DIRECT	0.535 [0.494, 0.579]	0.533	0.383	0.447	0.891	0.736
	R1-AFFORD	0.565 [0.519, 0.608]	0.566	0.428	0.325	0.829	0.695
	R2-TYPED	0.721 [0.679, 0.762]	0.711	0.549	0.120	0.853	0.764
	R3-SCALAR	0.446 [0.400, 0.487]	0.331	0.143	0.268	0.860	0.402
	R4-RECOG	0.608 [0.565, 0.652]	0.590	0.313	0.157	0.736	0.670
GEMINI-2.5-FLASH	R0-DIRECT	0.515 [0.471, 0.558]	0.500	0.273	0.536	0.938	0.778
	R1-AFFORD	0.594 [0.550, 0.637]	0.587	0.373	0.393	0.930	0.775
	R2-TYPED	0.652 [0.610, 0.694]	0.629	0.364	0.128	0.744	0.709
	R3-SCALAR	0.427 [0.381, 0.473]	0.297	0.000	0.208	0.783	0.374
	R4-RECOG	0.558 [0.512, 0.602]	0.513	0.374	0.282	0.860	0.623
LLAMA-3.3-70B	R0-DIRECT	0.492 [0.448, 0.537]	0.470	0.180	0.578	0.953	0.764
	R1-AFFORD	0.546 [0.500, 0.590]	0.535	0.372	0.447	0.938	0.727
	R2-TYPED	0.610 [0.565, 0.652]	0.592	0.326	0.094	0.674	0.650
	R3-SCALAR	0.452 [0.408, 0.496]	0.316	0.042	0.242	0.907	0.383
	R4-RECOG	0.613 [0.567, 0.658]	0.608	0.525	0.111	0.597	0.702
GPT-5	R0-DIRECT	0.521 [0.475, 0.565]	0.516	0.450	0.473	0.915	0.781
	R1-AFFORD	0.465 [0.421, 0.506]	0.468	0.425	0.427	0.837	0.732
	R2-TYPED	0.598 [0.554, 0.640]	0.610	0.490	0.254	0.775	0.748
	R3-SCALAR	0.417 [0.373, 0.465]	0.388	0.297	0.276	0.775	0.403
	R4-RECOG	0.552 [0.506, 0.598]	0.535	0.206	0.205	0.760	0.630

Table 3: Four-way routing on the CARB test split ($n=480$). Best per model in **bold**. Reference points: uniform random 0.231, best single-action policy 0.285. Typed prompting (R2-TYPED) is best or tied-best on accuracy for all four models and cuts over-commitment from 0.447–0.578 under a plain prompt to 0.094–0.254; routing an optimally-thresholded verbalised confidence (R3-SCALAR) is worst everywhere. Missing responses are scored as errors, which penalises GPT-5 (16–53 empty completions per regime; see section 5.3).

4.2 Experiment 1: recognition beats behaviour, but the recognition that fails is specific

Models know more than they do. Recognition beats default behaviour for all four models (+0.03 to +0.12 accuracy), significantly for LLAMA-3.3-70B ($p_{\text{Holm}} = 9.7 \times 10^{-4}$, $h = +0.24$) and marginally for CLAUDE-SONNET-4.5 ($p_{\text{Holm}} = 0.060$). The effect is far clearer on the metric the gap is actually about (figure 2): **over-commitment falls from 0.447–0.578 under a plain prompt to 0.111–0.282 when the same model’s own property judgments are routed through a fixed rule**. The models are not missing the information; they are not acting on it.

Naming the options is not enough. Adding a bare ask-affordance (R1-AFFORD) helps GEMINI-2.5-FLASH (+0.079, $p_{\text{Holm}} = 1.4 \times 10^{-4}$) and LLAMA-3.3-70B (+0.054, $p_{\text{Holm}} = 0.031$), does nothing for CLAUDE-SONNET-4.5 (+0.030, n.s.), and appears to hurt GPT-5 (−0.056, $p_{\text{Holm}} = 0.008$). That last result does not survive scrutiny: GPT-5 returned an empty completion on 53/480 R1-AFFORD items as reasoning tokens exhausted `max_tokens`, which the headline scoring counts as errors. Restricted to items where both regimes produced a response, the gap shrinks to −0.023 ($n = 424$, $p = 0.15$, uncorrected). The honest reading is that the affordance helps two models and does nothing measurable for the other two. Listing the options is not the intervention; *defining* them is (section 4.3).

Which recognition fails. Scoring each property judgment only where the decision rule reads it (table 4) gives the sharpest disconfirmation of a naive reading of the hypothesis. Models judge “is this safe and determinate?” at 0.893–0.957 and “is this within my capability?” at 0.943, but “was I told enough?” at **0.565** on the items that needed a question, and “can the user fix it in one turn?” at 0.574. **What models reliably notice is danger and their own limits, not underspecification** — which is exactly the property the hypothesis is about. The DEFER rows carry a second finding: on requests blocked by the model’s own capability, models call the request *not determinate* 37% of the time. They misattribute their own limitation to the world.

4.3 Experiment 2: typed structure beats scalar uncertainty everywhere

The head-to-head. Typed prompting beats optimally-routed scalar confidence on every model, by +0.181 (GPT-5), +0.275 (CLAUDE-SONNET-4.5), +0.225 (GEMINI-2.5-FLASH), and +0.158 (LLAMA-3.3-70B) accuracy; Cohen’s $h = 0.36, 0.57, 0.46, 0.32$; all $p_{\text{Holm}} < 10^{-8}$ (figure 3, table 3). Typed-deferral accuracy — getting the right *kind* of withholding — is 0.650–0.764 under typed prompting and 0.374–0.403 under the scalar. This replicates Unlu [2026] in direction at $15 \times$ the sample size and across five models, and corrects the magnitude: their 91.7% typed-deferral

Gold action	Property the rule reads	Accuracy	n
ACT	safe_and_determinate	0.957	510
ACT	within_capability	0.943	509
REFUSE	safe_and_determinate	0.893	439
ASK	within_capability	0.850	545
ACT	information_sufficient	0.782	510
ASK	safe_and_determinate	0.754	545
DEFER	within_capability	0.741	401
DEFER	safe_and_determinate	0.633	403
ASK	user_can_resolve	0.574	545
ASK	information_sufficient	0.565	545

Table 4: Recognition, decomposed. Each R4-RECOG property is scored only on items where the fixed decision rule actually reads it, pooled over the four models. Models are near-ceiling at “is this dangerous?” (0.893–0.957) and “can I even do this?” (0.943), and close to a coin flip at “was I told enough?” on the items that needed a question (0.565, bottom rows) — which is precisely the recognition the hypothesis is about. The DEFER rows are the second story: on requests blocked by the model’s own limits, models call the request indeterminate 37% of the time, misattributing their own limitation to the world.

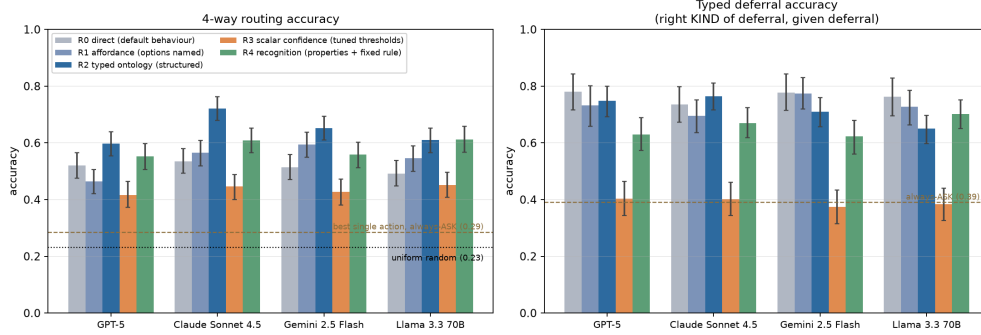


Figure 3: Four-way accuracy by elicitation regime on identical items ($n=480$), with bootstrap 95% CIs. Typed prompting (R2-TYPED) is the top or joint-top regime for every model; routing a verbalised confidence (R3-SCALAR) is the worst for every model, despite being given the best cross-validated router that can be fitted to it. Dashed lines mark uniform random (0.231) and the best single-action policy (0.285).

figure was an upper bound produced by a single-context audit, and with contexts isolated we see 0.65–0.76.

Why the scalar cannot work. Table 5 gives the geometric reason. The confidence number separates act from not-act well (AUROC 0.788–0.873) and is **at chance at separating DEFER from REFUSE** (0.436–0.563, below chance for CLAUDE-SONNET-4.5). Worse, the models do not order the two consistently: GPT-5 and CLAUDE-SONNET-4.5 put DEFER below REFUSE in mean confidence, GEMINI-2.5-FLASH and LLAMA-3.3-70B above it. Three distinct reasons for not acting cannot be recovered from an ordering of one number, and empirically they are not even consistently ordered.

Is the scalar at least complementary? No. A hybrid policy that uses the cross-validated scalar for the ACT gate — its one strength — and the typed judgment for the kind never beats plain typed prompting, and for CLAUDE-SONNET-4.5 it is significantly *worse* (0.654 vs. 0.721, $p_{\text{Holm}} = 1.1 \times 10^{-4}$). The same holds with a recognition-based second stage. The scalar adds nothing the typed judgment does not already contain.

Calibration and resolution. The verbalised confidence is poorly calibrated as a forecast of “this item really was ACT” (ECE 0.162–0.210 against a 0.27 base rate, mean stated confidence 0.364–0.446 — systematically over-confident). It is also barely a scale: **only 3–31% of the probability mass falls strictly between 0.05 and 0.95** (GEMINI-2.5-FLASH 3%, CLAUDE-SONNET-4.5 17%,

Model	AUROC	Mean conf.	AUROC, pairs of withholding types		
	ACT vs. rest	ACT/ASK/REFUSE/DEFER	ASK-REFUSE	ASK-DEFER	DEFER-REFUSE
GPT-5	0.814	83.3 / 50.8 / 23.6 / 11.0	0.721	0.780	0.507
CLAUDE-SONNET-4.5	0.824	82.7 / 45.2 / 18.9 / 6.8	0.669	0.741	0.436
GEMINI-2.5-FLASH	0.788	77.8 / 38.3 / 6.6 / 14.0	0.663	0.622	0.543
LLAMA-3.3-70B	0.873	84.0 / 34.5 / 9.6 / 19.5	0.652	0.596	0.563

Table 5: Diagnostics for the verbalised-confidence regime (R3-SCALAR). The scalar carries real information about *whether* to act (AUROC 0.79–0.87) and essentially none about *which kind* of withholding is right: DEFER vs. REFUSE is at chance for every model and below chance for CLAUDE-SONNET-4.5 (bold column). The mean-confidence column shows the models do not even order the two consistently — GPT-5 and CLAUDE-SONNET-4.5 place DEFER below REFUSE, GEMINI-2.5-FLASH and LLAMA-3.3-70B above it. Three deferral types cannot be recovered from an ordering of one number.

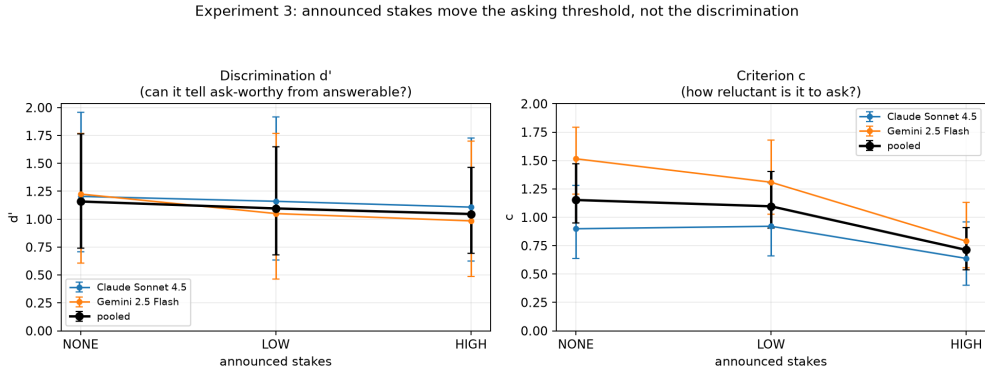


Figure 4: Announced stakes in signal-detection terms. Moving from NONE to HIGH slides the criterion c down decisively in both regimes while leaving d' flat under the typed ontology and *lowering* it under a plain ask-affordance. The manipulation buys indiscriminate caution, not calibrated caution.

GPT-5 18%, LLAMA-3.3-70B 31%). Asked for a number from 0 to 100, the models answer 0 or 100. Under an interaction budget, ranking by 1–confidence and asking on the least confident $b\%$ recovers roughly twice random ask-recall (0.33–0.39 at $b = 20\%$ against 0.20 random) — real but weak, and the typed policies sit at similar recall with much better precision (CLAUDE-SONNET-4.5: ask-rate 0.24, recall 0.41, precision 0.89).

4.4 Experiment 3: stakes move the threshold, not the judgment

Each of 240 items (60 per gold action) was shown under three announced-stakes frames prepended to the system prompt, within-item. We exclude GPT-5: its stakes cells came back 68–100% empty when the API budget ran out, and missingness that asymmetric across the manipulated factor cannot enter a factorial. The analysis pools CLAUDE-SONNET-4.5 and GEMINI-2.5-FLASH.

The manipulation works, in the wrong way. Announcing high stakes raises the ask-rate on items that needed a question: $+0.133$ typed ($p_{\text{Holm}} = 0.005$) and $+0.208$ free-text ($p_{\text{Holm}} = 1.4 \times 10^{-4}$), both paired HIGH vs. LOW. But it raises the false-alarm rate on genuinely answerable items as much or more (0.042 $\rightarrow$ 0.108 typed, 0.042 $\rightarrow$ 0.283 free-text). A paired bootstrap of the *change* between frames (table 6, figure 4) gives $\Delta c = -0.440$ [$-0.687, -0.274$], $p < 0.001$ with $\Delta d' = -0.112$ [$-0.623, +0.225$], $p = 0.53$ under the typed ontology. Under a plain ask-affordance — which is what a deployed assistant actually has — discrimination gets **significantly worse**: $\Delta d' = -0.581$ [$-1.205, -0.109$], $p = 0.013$.

Additive, not interactive. An item-clustered logistic regression of $P(\text{ASK})$ on ambiguity and announced stakes says the same thing in one number: ambiguous $\beta = +2.553$ ($p = 4.3 \times 10^{-9}$) and high $\beta = +0.630$ ($p = 0.009$) are both large, and the interaction ambiguous $\times$ high is

Regime	Frame	Hit rate (ask gold ASK)	False alarm (ask gold ACT)	d' [95% CI]	Criterion c [95% CI]
R2-TYPED (typed)	NONE	0.283	0.042	1.159 [0.74, 1.77]	1.152 [0.95, 1.47]
	LOW	0.292	0.050	1.096 [0.68, 1.65]	1.097 [0.90, 1.40]
	HIGH	0.425	0.108	1.046 [0.70, 1.46]	0.712 [0.54, 0.91]
R1-AFFORD (free text)	NONE	0.333	0.042	1.301 [0.91, 1.94]	1.081 [0.88, 1.42]
	LOW	0.350	0.058	1.184 [0.81, 1.70]	0.977 [0.79, 1.26]
	HIGH	0.558	0.283	0.720 [0.41, 1.06]	0.213 [0.06, 0.39]
<i>Paired bootstrap of the change, HIGH – NONE (4,000 resamples over items)</i>					
R2-TYPED (typed)		$\Delta d' = -0.112 [-0.623, +0.225], p = 0.53$		$\Delta c = -0.440 [-0.687, -0.274], p < 0.001$	
R1-AFFORD (free text)		$\Delta d' = -0.581 [-1.205, -0.109], p = 0.013$		$\Delta c = -0.868 [-1.202, -0.651], p < 0.001$	

Table 6: Announced stakes, decomposed by signal detection theory over 240 items shown under three frames within-item, pooled over CLAUDE-SONNET-4.5 and GEMINI-2.5-FLASH. Announcing high stakes moves the *criterion* decisively in both regimes and leaves *discrimination* statistically unchanged under the typed ontology — and makes it significantly worse under a plain ask-affordance, where the false-alarm rate rises almost sevenfold. Telling a model the stakes are high makes it more anxious, not a better judge.

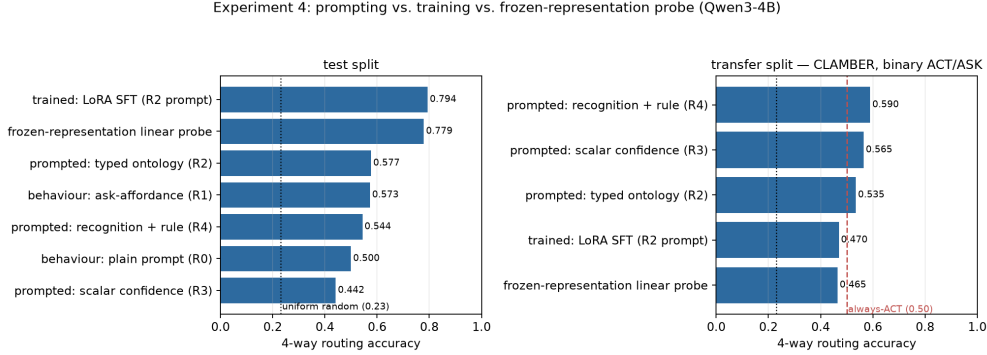


Figure 5: Prompting, training, and probing on QWEN3-4B. LoRA SFT reaches 0.794 and cuts over-commitment to 1.4%, but a frozen linear probe on the *base* model’s hidden state at the final prompt token gets within 1.5 points of it with no training at all — 20 points above what the same model does when simply asked. On the out-of-source transfer split both supervised conditions fall below the untrained prompted baselines.

$\beta = +0.010, p = 0.97$. A model that used stakes properly would show a positive interaction — asking *more where it matters*, not everywhere.

The cost is concrete. Under HIGH, the act-rate on genuinely answerable requests falls from 0.842 to 0.708 (typed) and from 0.925 to 0.550 (free text). Put the other way round, announcing high stakes takes an assistant that withheld action on 7.5% of answerable requests and turns it into one that withholds on **45%** of them.

Item-level stakes, where nobody tells the model anything. Among IN3 items whose gold action is ASK, the ask-rate rises with the annotated importance of the missing detail: 0.045 at importance 2 vs. 0.306 at importance 3 under typed prompting (item-level Mann–Whitney $p = 0.006, n = 22$ vs. 9). This is the one place where behaviour tracks stakes without being told. It is also confounded — a more important missing detail is plausibly also a more obvious one — so we treat it as suggestive rather than as evidence for the stakes clause.

4.5 Experiment 4: training helps in-distribution, and the signal was already there

The frontier patterns reproduce on an open model. QWEN3-4B shows both effects on its own (table 7): typed prompting (0.577) beats scalar routing (0.442) by 13.5 points on identical items, the same direction and roughly the same size as the four frontier models, and under a plain prompt it acts on **58%** of items it should have withheld action on, which the typed prompt cuts to 17% and the recognition rule to 13%. Whatever this is, it is not an artefact of one lab’s post-training.

Condition	Test acc [95% CI]	Macro-F1	ASK-F1	Over-commit ↓	Transfer acc
<i>Behaviour</i>					
plain prompt (R0-DIRECT)	0.500 [0.456, 0.546]	0.481	0.200	0.581	—
ask-affordance (R1-AFFORD)	0.573 [0.529, 0.617]	0.568	0.383	0.413	—
<i>Prompted</i>					
typed ontology (R2-TYPED)	0.577 [0.533, 0.621]	0.577	0.489	0.168	0.535
scalar confidence (R3-SCALAR)	0.442 [0.398, 0.483]	0.358	0.310	0.171	0.565
recognition + rule (R4-RECOG)	0.544 [0.500, 0.590]	0.539	0.530	0.131	0.590
<i>Supervised</i>					
LoRA SFT on typed routing	0.794 [0.756, 0.829]	0.790	0.798	0.014	0.470
frozen linear probe (base model)	0.779 [0.744, 0.817]	0.773	0.766	0.009	0.465

Table 7: Prompting, training, and probing on one open-weight model (QWEN3-4B). The open model reproduces both frontier patterns on its own: typed prompting beats scalar routing by 13.5 points on identical items, and a plain prompt over-commits on 58% of items that should have been withheld. LoRA SFT reaches **0.794** in-distribution (+0.217 over the same model with the same ontology, McNemar $p = 8.0 \times 10^{-15}$), but a frozen linear probe on the *untrained* model’s hidden state already reaches 0.779 — the routing signal was in the representation, and training mostly added a read-out. Neither survives a change of source: on the transfer split both fall below the untrained prompted baselines, and every condition sits within ± 0.09 of the always-ACT policy at 0.50.

Training works, in-distribution. LoRA SFT on 2,576 prompt→action pairs reaches **0.794**, +0.217 over the same model prompted with the same ontology (McNemar $p = 8.0 \times 10^{-15}$, $b=145 / c=41$), with over-commitment down to 0.014 and ASK-F1 up from 0.489 to 0.798 — in under two hours on one shared A6000.

But the signal was already in the representation. A logistic regression on the last hidden state at the final prompt token of the *base* model — no fine-tuning, no generation — reaches 0.779, **20 points above what the same model does when asked** (McNemar $p = 6.3 \times 10^{-12}$), within 1.5 points of the fine-tuned model. Read together with the SFT result: LoRA training did not teach the model to route, it taught it to *read out* a routing signal its frozen representation already carried. That is the representation-level version of the recognition–behaviour gap in section 4.2.

Neither transfers. On the out-of-source split the SFT model drops to 0.470 and the probe to 0.465 — both *below* the untrained prompted baselines (0.535–0.590), with the differences not significant in either direction (McNemar $p = 0.11$ and 0.13). The blunter framing is worse: CLAMBER is binary, so the relevant reference is always-ACT at 0.50, and every condition — trained, probed, or prompted — sits within ± 0.09 of just answering everything. The TF-IDF control shows the same shape (0.702 in-source → 0.355 out-of-source). Nothing in this study demonstrates a routing policy that generalises across sources.

4.6 Where the errors go

Pooling over the four models, four patterns stand out.

The dominant default error is answering an under-specified request. Under a plain prompt, 390 of 548 ASK items (71%) are answered outright. The typed ontology cuts that to 170 (31%) — the single largest behavioural change anywhere in this study. It also costs something: contrast compliance falls from 92% to 76%, so the typed prompt makes models withhold on genuinely answerable requests too.

Clarify-versus-abstain is exactly the confusion Baan et al. [2026] named. Under typed prompting, 144 of 548 ASK items are routed to REFUSE— 26%, almost as many as are routed correctly (178). The model recognises “I should not just answer this” and then picks *decline* rather than *ask*, converting a recoverable interaction into a dead end. No scalar-confidence route touches this, because the scalar cannot tell REFUSE from DEFER, let alone ASK from REFUSE.

The scalar router collapses two categories into one, visibly. Figure 6 shows what collapse means concretely: the cross-validated router assigns its lowest-confidence bin to a single label and everything low-confidence lands there. For CLAUDE-SONNET-4.5 that bin is DEFER (DEFER recall

Row-normalised confusion matrices (test split, n=480)

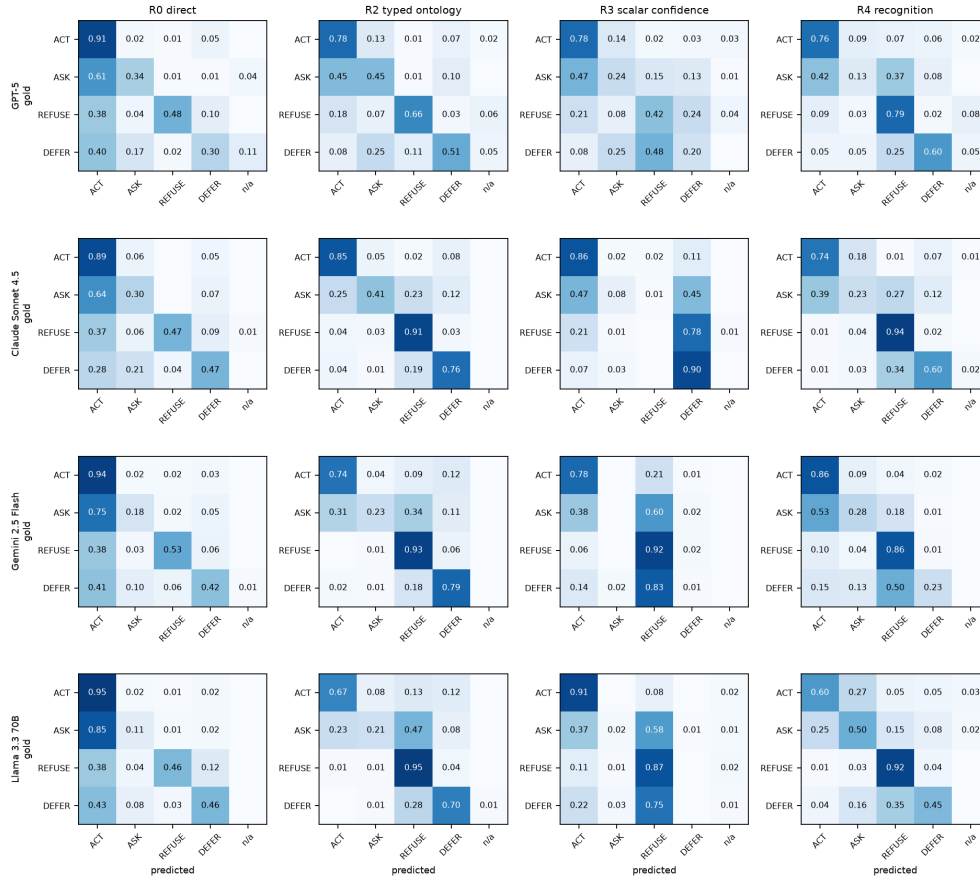


Figure 6: Confusion matrices by regime (rows = gold, columns = predicted). Two structures are visible. Under a plain prompt the mass sits in the ASK $\rightarrow$ ACT cell: models answer user-specified requests. Under the scalar router the mass collapses into a single low-confidence label — DEFER for CLAUDE-SONNET-4.5, REFUSE for GEMINI-2.5-FLASH and LLAMA-3.3-70B— so the router is making a two-way choice and labelling it three ways.

0.90, but 78% of REFUSE items sent to DEFER); for GEMINI-2.5-FLASH and LLAMA-3.3-70B it is REFUSE (REFUSE recall 0.92/0.87, but 83%/75% of DEFER items sent to REFUSE).

DEFER is systematically misread as REFUSE (77/408, 19%, under typed prompting), matching the property result that models judge their own capability blocks to be indeterminacies of the world. Qualitatively, the ASK $\rightarrow$ ACT errors are *plausible-completion* errors: given “Create a 30-minute at-home cardio workout plan”, the model’s own reason field reads “*The request is clear, safe, and within my capabilities*” — it substitutes a generic instantiation for the missing specifics and never registers that specifics were missing. The DEFER $\rightarrow$ REFUSE errors are attribution errors: for “a line-by-line analysis of every piece of legislation passed since 1789” the reason reads “*This request has no determinate scope*” rather than “this exceeds what I can produce”.

4.7 Is any of this worth deploying?

Accuracy prices every error the same, and the hypothesis is explicitly about stakes. We therefore score every policy under a cost model with one free parameter K , the cost of acting on a request that should have been withheld (correct action +1, needless question -0.3 , needless refusal -1.0 , withheld for the wrong reason $+0.2$).

Policy	$K=1$	$K=3$	$K=5$	$K=10$	$K=30$
oracle	+1.00	+1.00	+1.00	+1.00	+1.00
always ASK	+0.29	+0.29	+0.29	+0.29	+0.29
always ACT (today’s default)	−0.46	−1.93	−3.39	−7.04	−21.67
CLAUDE-SONNET-4.5, typed	+0.63	+0.46	+0.28	−0.15	−1.90
LLAMA-3.3-70B, typed	+0.51	+0.37	+0.23	−0.12	−1.54
GEMINI-2.5-FLASH, typed	+0.53	+0.35	+0.16	−0.31	−2.18
GPT-5, typed	+0.39	−0.03	−0.46	−1.51	−5.72
CLAUDE-SONNET-4.5, default behaviour	+0.21	−0.45	−1.11	−2.75	−9.34
GPT-5, default behaviour	+0.14	−0.62	−1.38	−3.27	−10.86
QWEN3-4B frozen probe	+0.69	+0.68	+0.67	+0.64	+0.51
QWEN3-4B LoRA SFT	+0.70	+0.68	+0.66	+0.61	+0.40

Table 8: Mean utility per request when errors are priced. K is the cost of acting on a request that should have been withheld; a correct action scores +1, a needless question −0.3, a needless refusal −1.0, and withholding for the wrong reason +0.2. Read down the columns: every frontier model’s best prompted policy crosses below the trivial *always*-ASK policy at $K \approx 5$ –10 and goes negative shortly after, and every model’s *default behaviour* goes negative between $K = 1$ and $K = 2$. The two supervised QWEN3-4B policies stay above *always*-ASK across the whole sweep — but table 7 shows neither survives a change of source, so that is a statement about CARB, not about deployment.

Table 8 and figure 7 give the deployment-relevant reading. Every model’s *default behaviour* goes negative between $K = 1$ and $K = 2$, i.e., as soon as a wrong action costs even slightly more than a needless refusal. Every frontier model’s best prompted policy crosses below the trivial *always*-ASK policy at $K \approx 5$ –10 and goes negative shortly after. Only the two in-distribution supervised QWEN3-4B policies stay above *always*-ASK across the sweep, and section 4.5 has already shown neither survives a change of source. The conclusion is not that these models cannot route; it is that **their routing is not yet good enough to beat asking every time** in exactly the regime the hypothesis cares about.

4.8 Sensitivity to the label mapping

Because the gold labels are inherited rather than authored, the mapping table is the main place a systematic error could enter. We recomputed every model $\times$ regime cell three ways: dropping the two contested mapping cells, under an alternative mapping that sends false-presupposition items to REFUSE, and under a capability re-audit of the IN3 items. Individual accuracies move by at most **0.068**, and **both headline orderings survive in all 16 model $\times$ variant cells**: typed beats scalar everywhere, and recognition beats default behaviour everywhere. The rank order is not completely invariant — for LLAMA-3.3-70B the pre-registered labels put recognition 0.003 above typed, which every variant reverses — but no conclusion above depends on that pair.

5 Discussion

5.1 What the results say about the hypothesis, clause by clause

The hypothesis we set out to test has five separable claims, and the design lets each fail alone.

“Can be prompted to recognise when it lacks sufficient information” — partly true, and weaker than the literature suggests. Recognition beats behaviour for all four models, replicating Su and Cardie [2026]. But recognition of the specific property at issue is 0.565 on the items where it matters, against 0.893–0.957 for safety and 0.943 for capability. The blanket statement “models recognise ambiguity” is too generous: what they reliably recognise is danger and their own limits. Underspecification is the blind spot, and it is the one the hypothesis names.

“By estimating its uncertainty” — false as a mechanism. This is the clearest result in the paper. An optimally-routed scalar confidence loses to typed prompting by 0.158–0.275 accuracy on every model, is at chance at distinguishing the three reasons for not acting, has almost no resolution to threshold (3–31% of mass strictly between 0.05 and 0.95), and adds nothing when bolted onto a

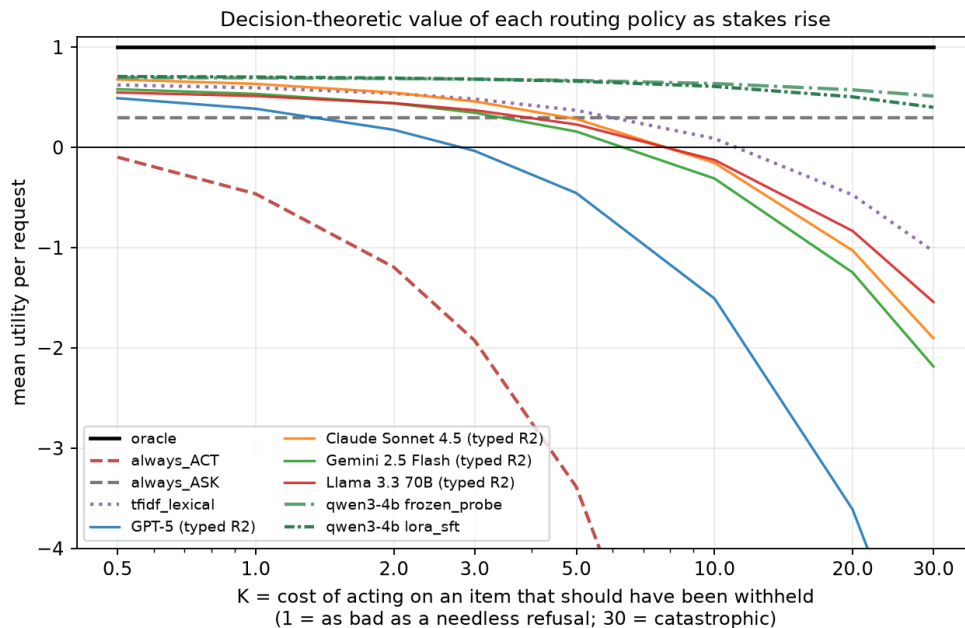


Figure 7: Mean utility per request as the cost K of an unwarranted action rises. The flat always-ASK line is the bar that matters: every frontier model’s best prompted policy crosses below it at $K \approx 5$ –10. $K = 10$ is not extreme — it is “a wrong action costs ten times a needless refusal”, about $33 \times$ a needless clarifying question, which is mild for anything that writes to a database, sends a message, or gives medical advice.

typed policy. The failure is not that the estimator is imprecise — it is that the target has the wrong dimensionality. What works is a *structural* intervention: defining the action space. This is where the entire measurable gain lives.

“Choosing among acting, asking, refusing, deferring” — true, once the ontology is given. 0.598–0.721 four-way accuracy against 0.231 random and 0.285 best-single-action, with over-commitment down from 45–58% to 9–25%.

“Depending on the stakes” — false in the sense that matters. Announced stakes move the criterion ($\Delta c = -0.44$, $p < 0.001$) and not the discrimination ($\Delta d' = -0.11$, $p = 0.53$); under a plain ask-affordance they make discrimination significantly worse; and the ambiguity $\times$ stakes interaction is exactly zero ($\beta = +0.010$, $p = 0.97$). A system-prompt declaration of high stakes buys indiscriminate caution at a real price in needless refusals.

“Can be trained” — true in-distribution, unsupported out of it. LoRA SFT lifts typed routing from 0.577 to 0.794 and drops over-commitment to 1.4%. But it scores 0.470 out-of-source, below its own untrained prompted baseline, and a frozen probe on the base model already reaches 0.779 with no training. Most of what SFT delivered was reading out a signal the representation already carried, and that read-out is partly source-specific. Since the practical value of an ask-policy lies entirely in generalising to requests unlike the ones it was tuned on, this is the weaker of the two readings.

5.2 Two mechanisms worth naming

The gap is a read-out gap, at two levels. At the behavioural level, a model’s own property judgments routed through a fixed rule beat what it does when simply asked. At the representational level, a linear probe on the base model’s frozen hidden state beats what the same model does when asked, by 20 points. Both say the same thing: the information needed to route is present and is not being used. That reframes the intervention target. Better uncertainty estimation adds signal that is already there; what is missing is the step from an internal state to a choice of action.

Deferral is mis-attributed, not mis-detected. Models do not fail to notice that a request is blocked — they fail to locate the blocker. On requests blocked by the model’s own capability, they call the request indeterminate 37% of the time, and 19% of DEFER items are routed to REFUSE. The user-facing consequence is specific and bad: told “that is impossible” when the truth is “I cannot do that, but someone else can”, a user abandons a task that was achievable.

5.3 Limitations

The judge is a model. Free-text labels come from QWEN3-14B, not a human. Two checks put it in the substantial-agreement band ($\kappa = 0.781$ against an independent second judge over 3,768 items, $\kappa = 0.750$ against 80 blind annotations), but the blind annotator was itself a model, so this is a third model agreeing rather than human validation. The judge also has a known REFUSE/DEFER confusion — the same distinction the models find hard — so the absolute levels in Experiment 1 are softer than its within-item contrasts.

Lexical shortcuts inflate absolute accuracy. TF-IDF reaches 0.702 in-source (section 4.1). We report this rather than working around it, and restrict every headline claim to within-item contrasts between regimes, which the shortcut affects equally on both sides.

Missing data penalises GPT-5. Its completions were truncated on 3–11% of main-run items as reasoning tokens exhausted `max_tokens`, and on 56–100% of its stakes cells when the API budget ran out. Headline numbers score a missing response as an error — the deployment-realistic reading — and we report complete-case sensitivity per comparison. GPT-5 is excluded from Experiment 3 entirely.

Stakes are announced, not experienced. No dataset in our corpus encodes consequence-based stakes, so Experiment 3 manipulates *stated* stakes in the system prompt and its item-level arm uses annotated *importance*. A model that ignores announced stakes might still behave well under real ones. This is the single largest gap between what we measured and what the hypothesis claims.

One benchmark, one language, single-turn, one out-of-source check. CARB is English, single-turn, and dominated by COCONOT. The only out-of-source split, CLAMBER, is binary and therefore cannot test the deferral-type distinction the study is about; on it, every condition sits within ± 0.09 of always-ACT. So the strongest claims here — structure beats scalars, recognition beats behaviour — are demonstrated *within* one source family, and the one test of cross-source generalisation is both weak and negative. We measure whether to ask; *what* to ask and *when to stop asking* are out of scope.

Reproducibility, checked rather than asserted. Every deterministic analysis was re-run from the stored raw outputs and produced byte-identical JSON, and every quantitative claim in this paper is restated as an executable assertion that re-checks it against the JSON that produced it (42 assertions, non-zero exit if any number drifts). What is *not* reproducible from scratch is the raw data: the API budget is spent and providers do not guarantee identical completions at temperature 0. We ship the response cache so the analysis pipeline can be re-run end to end.

5.4 Implications for building systems

Three recommendations follow directly. **Do not ask a model for a confidence number and threshold it** — give it the action space explicitly, with definitions and an ordered decision procedure, which is where the entire measurable gain lives. **Do not expect a “this is high-stakes” system prompt to buy calibrated caution;** it buys indiscriminate caution, and under a plain ask-affordance it degrades discrimination outright. **Benchmark any router against always asking**, because below a surprisingly low error cost that trivial policy is the better deal, and accuracy hides this completely.

There is also a broader-impact reading of the utility result that cuts both ways. An assistant tuned to ask more will interrupt users more, and the free-text stakes condition shows how fast that degrades: withholding action on 45% of perfectly answerable requests is its own failure mode, and one that falls hardest on users whose requests only *look* unusual. Routing policies should be evaluated on contrast sets, not only on the items where asking is right.

6 Conclusion

We asked whether a model can tell when it lacks the information to act safely, and choose among acting, asking, refusing, and deferring. To answer it we built CARB, an 840-item four-way action-routing benchmark with source-inherited labels, and held the items fixed while varying only how the decision is elicited, across five regimes and five models.

The answer is a qualified yes with a specific correction. Models can be prompted into a usable four-way policy — 0.598–0.721 accuracy against 0.231 random, with over-commitment falling from 45–58% to 9–25% — but not by the mechanism the hypothesis names. A verbalised confidence, given the best router that can be fitted to it, loses to typed prompting by 0.158–0.275 on every model and is at chance at telling apart the three reasons for not acting. The recognition that fails is specific: models judge safety at 0.893 and their own capability at 0.943, and information sufficiency at 0.565. Announced stakes move the decision criterion and not the discrimination, with an ambiguity $\times$ stakes interaction of exactly zero. Fine-tuning an open 4B model reaches 0.794 in-distribution and 0.470 out of it, where a frozen linear probe on the untrained model already reaches 0.779 — so training mostly supplied a read-out for a signal the representation already carried.

The takeaway: the bottleneck in deciding whether to ask is structural, not statistical. Give a model the action space and it routes; give it a confidence scale and it cannot, because three reasons for not acting do not fit on one axis. And until a routing policy beats *always asking* at a realistic error cost, the honest baseline for any such system is the trivial one.

Future work, in priority order. (1) A stakes manipulation with *experienced* consequences — an executable task where a wrong action actually fails a test — since announced stakes are the weakest link here. (2) Training directly on the ASK-F1 objective rather than the typed label, following the transfer result of Trinh et al. [2026]. (3) Probing what the frozen representation decodes that does not survive a source change, which is the crux of the negative transfer result. (4) Multi-turn: whether to ask is not the same decision as when to stop.

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A Additional details

A.1 Deviations from the pre-registered plan

We record every deviation. (i) The behavioural judge is a local QWEN3-14B model rather than a hosted GPT-4.1: the API budget was exhausted *after* all raw completions were collected but *before* judging, so every hosted judge call returned HTTP 403. The failed attempt is kept in the artefacts for provenance. (ii) The scalar regime’s thresholds are 5-fold cross-validated on the test split rather than fitted on dev, because the dev split was never collected before the budget ran out; cross-validation is the stricter of the two choices, since no item is predicted by a router that saw it. (iii) The LoRA run is 2 epochs at batch $4 \times$ accumulation 4, not $3 \times 8 \times 2$: the first attempt was OOM-killed by a co-tenant process on the shared GPU, and validation loss had already converged after epoch 0 in both attempts (0.0129 and 0.0095; the completed run’s epoch-1 value is 0.0069). (iv) IN3’s importance field failed to parse in the original build — the column ships as a list, not a repr-string — and was recovered before analysis. Only the previously-null importance and derived stakes fields changed; no gold labels moved.

A.2 Complete-case sensitivity for GPT-5

GPT-5 returned empty completions on 16/480 (R0-DIRECT), 53/480 (R1-AFFORD), 14/480 (R2-TYPED), 10/480 (R3-SCALAR), and 16/480 (R4-RECOG) test items, in every case because reasoning tokens exhausted `max_tokens`. The headline scoring counts these as errors, which is the deployment-realistic reading — a response the user never sees is not a correct action. Under complete-case scoring the one comparison whose sign changes is the affordance effect: -0.056 ($p_{\text{Holm}} = 0.008$) in the headline becomes -0.023 ($n = 424$, $p = 0.15$, uncorrected). No other conclusion in the paper is affected.

A.3 Cost model for the utility analysis

Let K be the cost of acting on a request that should have been withheld. A correct action scores +1; a needless clarifying question on an answerable request scores -0.3 ; a needless refusal scores -1.0 ; withholding action for the wrong reason (e.g., refusing when deferring was right) scores $+0.2$, since the immediate harm is avoided but the interaction is degraded. Utilities are means over the 480 test items. The parameter K is the only free quantity, and the qualitative conclusion — that prompted frontier policies cross below always-ASK at moderate K — holds for any assignment in which a needless question is cheaper than a needless refusal.

A.4 Artefacts

The release contains the benchmark (840 items plus the 2,800-item training pool and the published label-mapping table), 9,600 raw completions from the main run, 4,320 from the stakes run, all judge labels from both judges, the LoRA adapter and training curve, every analysis JSON, all ten figures, and a claim-checking script that re-verifies 42 quantitative assertions against the JSON that produced them.